

Chapter 3: Labeling in Financial Machine Learning

Based on *Advances in Financial Machine Learning* by Marcos M. López de Prado

AI Finance Seminar

Sogang Univ

September 2026

Chapter Outline

- 1 Motivation & Supervised Learning in Finance
- 2 The Fixed-Time Horizon Method & Its Pitfalls
- 3 Dynamic Thresholds
- 4 The Triple-Barrier Method
- 5 Learning Side vs. Size
- 6 Meta-Labeling & The Quantamental Approach
- 7 Handling Class Imbalance
- 8 Summary & Key Takeaways

Motivation: The Labeling Challenge in Finance

Supervised Learning Requirement

Supervised ML algorithms require a feature matrix X paired with a target array y . While feature engineering (X) extracts signals, **labeling** (y) defines the exact task the algorithm learns.

- **Financial Nuisances:** Financial time series exhibit low signal-to-noise ratio, non-stationarity, heteroscedasticity, and path-dependent risk.
- **A Critical Mistake:** Naive labeling methods from non-financial ML fail because they ignore trading mechanics (stop-loss, profit-taking) and market volatility regimes.
- **Chapter Objective:** Move from fixed-time horizon benchmarks to **path-dependent triple-barrier labeling** and **meta-labeling**.

The Fixed-Time Horizon Method

Standard Academic Formulation

Given bar index $t_{i,0}$ and a fixed horizon of h bars, return is defined as:

$$r_{t_{i,0}, t_{i,0}+h} = \frac{p_{t_{i,0}+h}}{p_{t_{i,0}}} - 1$$

An observation X_i is assigned label $y_i \in \{-1, 0, 1\}$ using a constant threshold τ :

$$y_i = \begin{cases} -1 & \text{if } r_{t_{i,0}, t_{i,0}+h} < -\tau \\ 0 & \text{if } |r_{t_{i,0}, t_{i,0}+h}| \leq \tau \\ 1 & \text{if } r_{t_{i,0}, t_{i,0}+h} > \tau \end{cases}$$

Why Fixed-Time Horizon Fails in Practice

1. Heteroscedasticity & Seasonality

- Volatility fluctuates wildly across intraday sessions (e.g., market open vs. noon).
- Applying a constant threshold τ causes label distribution to inherit seasonality: high volatility periods trigger extreme labels; low volatility periods generate mostly 0s.

2. Path-Ignorance

- Ignores price path $[t_{i,0}, t_{i,0} + h]$.
- A position might hit an exchange stop-loss or margin call at $t_{i,0} + \Delta t$, yet be labeled a "profit" if price recovers by $t_{i,0} + h$.
- Unrealistic: Real portfolios cannot survive intermediate liquidation.

Computing Dynamic Volatility Thresholds

Adaptive Volatility Thresholds ($\sigma_{t_{i,0}}$)

Instead of a fixed constant τ , thresholds should adjust dynamically based on prevailing market volatility $\sigma_{t_{i,0}}$.

- **EWMA Volatility Estimation:** Compute a rolling Exponentially Weighted Moving Standard Deviation of returns over a span span_0 (e.g., 100 days):

$$\sigma_{t_{i,0}} = \text{EWMA}(r_t, \text{span}_0)$$

- **Standardized Labels:** Scale profit-taking and stop-loss barriers directly by $\sigma_{t_{i,0}}$.
- **Benefit:** Prevents label clustering during volatile regimes and restores statistical stationarity across market cycles.

The Triple-Barrier Method: Concept

Simulates realistic portfolio trading rules by setting **three simultaneous exit barriers**:

- 1 **Upper Horizontal Barrier (Profit-Taking)**: Triggered when unrealized profit reaches target $pt \cdot \sigma_{t_{i,0}} \implies y_i = 1$.
- 2 **Lower Horizontal Barrier (Stop-Loss)**: Triggered when unrealized loss hits limit $-sl \cdot \sigma_{t_{i,0}} \implies y_i = -1$.
- 3 **Vertical Barrier (Expiration t_1)**: Triggered after a maximum holding period of h bars elapsed $\implies y_i = 0$ or $\text{sgn}(r_{t_{i,0}, t_{i,1}})$.

Path Dependency

The label y_i is determined by $t_{i,1} = \min\{\text{touch time of upper, lower, or vertical barrier}\}$, capturing the full price trajectory over $[t_{i,0}, t_{i,1}]$.

Triple-Barrier Configurations

Barriers are represented by the triplet $[pt, sl, t_1]$ where 1 = active and 0 = disabled:

Triplet $[pt, sl, t_1]$	Type	Practical Application
$[1, 1, 1]$	Standard Setup	Profit-taking, stop-loss, and max holding time.
$[0, 1, 1]$	Time Exit + Stop	Hold for h bars unless stopped out.
$[1, 1, 0]$	Unbounded Time	Hold until profit or loss target hit (no time limit).
$[0, 1, 0]$	Illogical	Hold aimlessly until stopped out.
$[0, 0, 0]$	Disabled	No barriers, no labels generated.

Learning Side and Size

Two Distinct ML Tasks

- **Learning Side (Primary Prediction):** Forecasting whether to go Long (+1) or Short (−1) without prior directional signals. Horizontal barriers must be ****symmetric**** because profit vs. loss orientation is unknown.
 - **Learning Size (Bet Sizing):** Determining how much capital to allocate ($0 \leq m \leq 1$) given an existing directional forecast.
-
- When side is known ex-ante, horizontal barriers can be ****asymmetric**** ($pt \neq sl$).
 - Decoupling Side from Size leads directly to ****Meta-Labeling****.

Meta-Labeling: The Secondary ML Model

Definition

Meta-Labeling trains a ****secondary ML model**** on the predictions of a primary model to decide whether to take the bet or pass ($y \in \{0, 1\}$).

Primary Model (Exogenous / White-Box):

- Determines position **Side** ($\text{side} \in \{-1, 1\}$).
- Examples: Economic theory, SMA crossover, fundamental analyst calls.
- High Recall, but lower Precision.

Secondary Model (ML Overlay):

- Determines position **Size / Action** ($y_i \in \{0, 1\}$).
- $y_i = 1$ if primary signal yields profit.
- $y_i = 0$ if primary signal fails (loss / timeout).

Advantages of Meta-Labeling

- ➊ **F1-Score & Precision Optimization:** Filters out False Positives without sacrificing true opportunities identified by the primary model.
- ➋ **Quantamental Integration:** Allows hedge funds to layer ML on top of white-box fundamental or discretionary models.
- ➌ **Protection Against Overfitting:** ML does not decide trade direction, only execution/bet size—dramatically reducing search space and overfit risk.
- ➍ **Asymmetric Strategy Design:** Enables separate long-only and short-only overlay models driven by distinct feature sets.
- ➎ **Prevents Sizing Catastrophes:** High accuracy on small bets and low accuracy on large bets leads to ruin; meta-labeling optimizes sizing probability.

Dropping Unnecessary / Rare Labels

Class Imbalance Problem

If a label class (e.g., 0 for horizontal vertical touches) occurs in $< 5\%$ of observations, ML classifiers suffer from degraded boundary estimation.

- **Solution ('dropLabels' Algorithm):** Recursively drop observations associated with extremely rare classes until remaining classes exceed a minimum percentage threshold minPct.
- **Result:** Focuses classification capacity on distinct, statistically meaningful market outcomes.

Summary: Labeling Rules in Financial ML

Key Takeaways from AFML Chapter 3

- **Avoid Fixed-Time Horizon:** Fixed time bars with constant thresholds create non-stationary, heteroscedastic labels that ignore path risk.
- **Use Dynamic Volatility Barriers:** Anchor profit-taking and stop-loss limits to rolling EWMA volatility $\sigma_{t_i,0}$.
- **Adopt the Triple-Barrier Method:** Model realistic path-dependent trading exits (profit-taking, stop-loss, holding time limit).
- **Leverage Meta-Labeling:** Separate side prediction from bet sizing to boost Precision, elevate F1-score, and integrate white-box models cleanly.